PYTHAGORAS, METALLIC RATIOS AND FIELD EXTENSIONS ON ONE PAGE

Metallic triangles and new square roots

A quadratic gives two legs. Pythagoras asks whether their hypotenuse needs a larger number system.

1 / MAKE A RATIO INTO TWO PERPENDICULAR LENGTHS

Choose a metallic length and its reciprocal

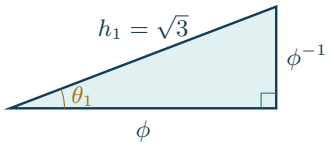
For integer $\lambda \geq 1$, take the positive root of $x^2 - \lambda x - 1 = 0$:

$$\mu_\lambda = \frac{\lambda + \sqrt{\lambda^2 + 4}}{2}, \qquad \mu_\lambda - \mu_\lambda^{-1} = \lambda.$$

Here a **metallic right triangle** has legs μ_λ and μ_λ^{-1}.
Their product is 1, so its area is always $1/2$. Pythagoras gives

$$\boxed{h_\lambda^2 = \mu_\lambda^2 + \mu_\lambda^{-2} = (\mu_\lambda - \mu_\lambda^{-1})^2 + 2 = \lambda^2 + 2.}$$

The acute angle beside the longer leg obeys $\tan\theta_\lambda = \mu_\lambda^{-2}$.



Golden case: $\phi = \mu_1$.
The legs are ϕ and $\phi - 1$.

2 / THE SAME CONSTRUCTION FOR EACH METAL

The legs come from one root; the hypotenuse may introduce another

λ	ratio	longer leg μ_λ	shorter leg μ_λ^{-1}	$h_\lambda = \sqrt{\lambda^2+2}$
1	Golden	$(1+\sqrt{5})/2$	$(\sqrt{5}-1)/2$	$\sqrt{3}$
2	Silver	$1+\sqrt{2}$	$\sqrt{2}-1$	$\sqrt{6}$
3	Bronze	$(3+\sqrt{13})/2$	$(\sqrt{13}-3)/2$	$\sqrt{11}$
4	4-metallic	$2+\sqrt{5}$	$\sqrt{5}-2$	$3\sqrt{2}$

3 / A RELATED FAMILY LABELLED BY A RADICAND

Fix the difference of the legs at one

For integer $d > 1$, set $A_d = (\sqrt{d}+1)/2$ and $B_d = (\sqrt{d}-1)/2$. Then $A_d - B_d = 1$, and the cross terms cancel:

$$H_d^2 = A_d^2 + B_d^2 = \frac{(\sqrt{d}+1)^2 + (\sqrt{d}-1)^2}{4} = \frac{d+1}{2}, \qquad \tan\theta_d = \frac{\sqrt{d}-1}{\sqrt{d}+1}.$$

d	hypotenuse H_d	$\tan\theta_d$	what this gives
2	$\sqrt{6}/2$	$3-2\sqrt{2}$	double all lengths to get silver
3	$\sqrt{2}$	$2-\sqrt{3}$	$\theta_3 = 15°$
5	$\sqrt{3}$	$(3-\sqrt{5})/2$	the golden right triangle
7	2	$(4-\sqrt{7})/3$	a rational hypotenuse

Here d is a radicand; λ is the metallic index.
The product $A_d B_d = (d-1)/4$ equals 1 at $d = 5$.
The two families meet there, and the $d = 2$ case is a scaled silver triangle.

4 / ALLOW THE ARITHMETIC A LITTLE MORE ROOM

One new root gives two number coordinates

Let $\mathbb{Q}$ denote the rational numbers. **Adjoining** $\sqrt{5}$ means allowing every number of the form

$$K = \mathbb{Q}(\sqrt{5}) = \{a + b\sqrt{5} : a, b \in \mathbb{Q}\}.$$

Addition and multiplication keep this form. Division works too, using the conjugate:

$$\frac{1}{a+b\sqrt{5}} = \frac{a-b\sqrt{5}}{a^2-5b^2}, \qquad a + b\sqrt{5} \neq 0.$$

Thus K is a **field**: the four arithmetic operations stay inside it, with division by zero excluded.
Every element has two unique rational coordinates, a, b, using the building blocks $1, \sqrt{5}$.
This is a **quadratic field extension** of $\mathbb{Q}$: its **degree** is 2.
Both golden legs already belong to this field.

5 / ASK WHAT ALL THREE LENGTHS REQUIRE

Pythagoras can ask for a second root

The golden legs use $\sqrt{5}$, but their hypotenuse is $\sqrt{3}$.
No rational a, b can write $\sqrt{3} = a + b\sqrt{5}$: squaring would force $ab = 0$, and neither case works. So enlarge the field again:

$$\mathbb{Q} \subset \mathbb{Q}(\sqrt{5}) \subset \mathbb{Q}(\sqrt{5}, \sqrt{3}).$$

The last field consists of

$$r + s\sqrt{5} + t\sqrt{3} + u\sqrt{15}, \qquad r, s, t, u \in \mathbb{Q}.$$

Multiplying the two roots creates $\sqrt{15}$.
There are now four independent rational coordinates: the extension has **degree** 4 **over** $\mathbb{Q}$.
For the $d = 7$ triangle, the hypotenuse is already 2, so $\mathbb{Q}(\sqrt{7})$ contains all three lengths.
Pythagoras does not require an extra root in that case.

Further reading. J. B. Gil and A. Worley, *Generalized Metallic Means*, §1; K. McGerty, *Algebra II: Rings and Modules*, §6. Companions: metallic ratios, continued fractions, and Gaussian integers.

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